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4.3.R1 (1/1 point)

Suppose we collect data for a group of students in a statistics class with variables X_1 = hours studied, X_2 = undergrad GPA, and Y = receive an A. We fit a logistic regression and produce estimated coefficients $\hat{\beta}_0 = -6, \hat{\beta}_1 = 0.05, \hat{\beta}_2 = 1$.

Estimate the probability that a student who studies for 40h and has an undergrad GPA of 3.5 gets an A in the class:

Answer: .3775**EXPLANATION**

We know that $P((40, 3.5)) = \frac{e^{-6 + .05 \cdot 40 + 1 \cdot 3.5}}{1 + e^{-6 + .05 \cdot 40 + 1 \cdot 3.5}} = .37754$

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4.3.R2 (1/1 point)

How many hours would that student need to study to have a 50% chance of getting an A in the class?:

Answer: 50**EXPLANATION**

We have $P((h, 3.5)) = \frac{e^{-6 + .05 \cdot h + 1 \cdot 3.5}}{1 + e^{-6 + .05 \cdot h + 1 \cdot 3.5}} = .5$. Rearranging gives $-6 + .05 \cdot h + 1 \cdot 3.5 = 0$ or $h = 50$

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